A FINITE EXTENSION IS GALOIS EXACTLY WHEN ITS AUTOMORPHISM GROUP HAS ORDER EQUAL TO THE DEGREE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file fields.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: This is the counting reformulation of Galois that the chapter uses to detect Galois extensions without exhibiting splittings.

Editorial note. The label lemma-finite-Galois names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Chapter lemma-normal-and-automorphisms (source lemma): for a finite extension E/F, |Aut(E/F)| <= [E:F]_s with equality if and only if E is normal over F
- Chapter fact around the separable degree definition: [E:F]_s = [E:F] exactly when E/F is separable
- Definition in this chapter: Galois means normal and separable

Source reference. Original reference 09HS: sources/stacks-fields/pinned-original-source.tex, source lines 2211–2211. Prerequisite pointer invoked at 2888; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let E/F be a finite extension of fields. Then E is Galois over F if and only if $|Aut(E/F)| = [E : F]$.*

Proof. Assume $|\mathrm{Aut}(E/F)| = [E : F]$. By Lemma 09HS this implies that E/F is separable and normal, hence Galois. Conversely, if E/F is separable then $[E : F] = [E : F]_s$ and if E/F is in addition normal, then Lemma 09HS implies that $|\mathrm{Aut}(E/F)| = [E : F]$. □

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